Team Green

CS 410

Spring 2026

Take the Lead

Have you ever struggled financially to pay a vet bill and your pet was left without treatment? Take the Lead will allow a 30 day campaign to run so the owner can generate funds so the pet can be seen immediately.

Meet the Team

Student

Siam Siam

My name is Siam, and I am a senior Computer Science student expecting to graduate Fall 2026. I was raised in Newport News, Virginia, and graduated from Warwick High School. Outside of school, I enjoy creating art, working out, fashion, and playing the guitar.

Student

Madeline Lacey

Computer Science student, exp 2026, I was born in Pasadena, California. I moved to Virginia Beach in 2017 and attended Tidewater Community College for an associates' degree and continued my education at Old Dominion University.

Student

Jordan Lopez

Computer Science student, Born in Hampton, Virginia and I am expected to graduate in the Fall 2026. Currently employed at Huntington Ingalls Industries as a software developer with project manager responsibilities.

Meet the Team

Student

Jaden Evans

I am Jaden Evans, a senior computer science major here at ODU and a Virginia Beach native. I graduated from Green Run Collegiate in 2022 and hope to graduate ODU this fall. I enjoy software development, cooking, and writing.

Student

Jermaine Lockett

Hi! I'm a Computer Science student who's currently working for the Systems Group as a Consultant. Outside of school I enjoy biking, cars, homelabbing, and cooking.

Student

Harrison Basil

A Computer Science student working for the Systems Group as a System Administrator.

Overview

Problem

- **Insurance**
  Approximately 0.06% of pet households are insured.

- **Lack of Information**
  Many pet parents have a lack of accessible, easily understandable information on their pet's health.

Opportunity

- **The Power of Empathy**
  GoFundMe-like campaigns have proven successful in raising funds for pets' needs.

- **Knowledge is Power**
  With modern technology, providing access to vetted information is easier than ever.

Solution

- **Give a Dime, Save a Pet**
  We provide an easy, safe, and direct form to donate to pets' vet expenses.

- **Ease-of-Access Info**
  We provide info and chat boxes on pets' health issues, insurance, hospitals, etc.

The Problem Statement

The care and maintenance of animals is an expensive venture with not a lot of information and financial assistance readily available.

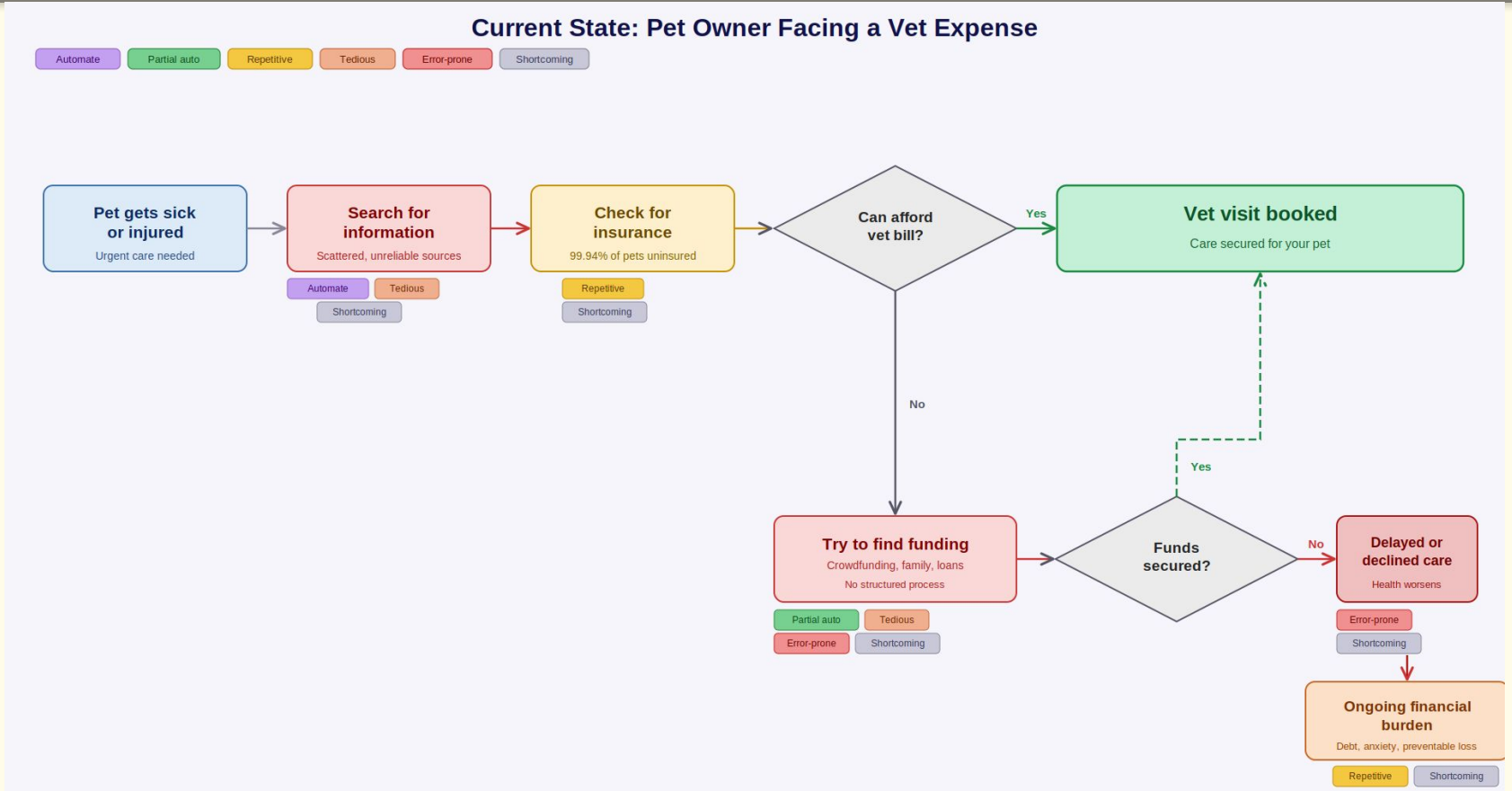
Problem Characteristics

There is a distinct lack of information available to pet owners on their options for their pets' health

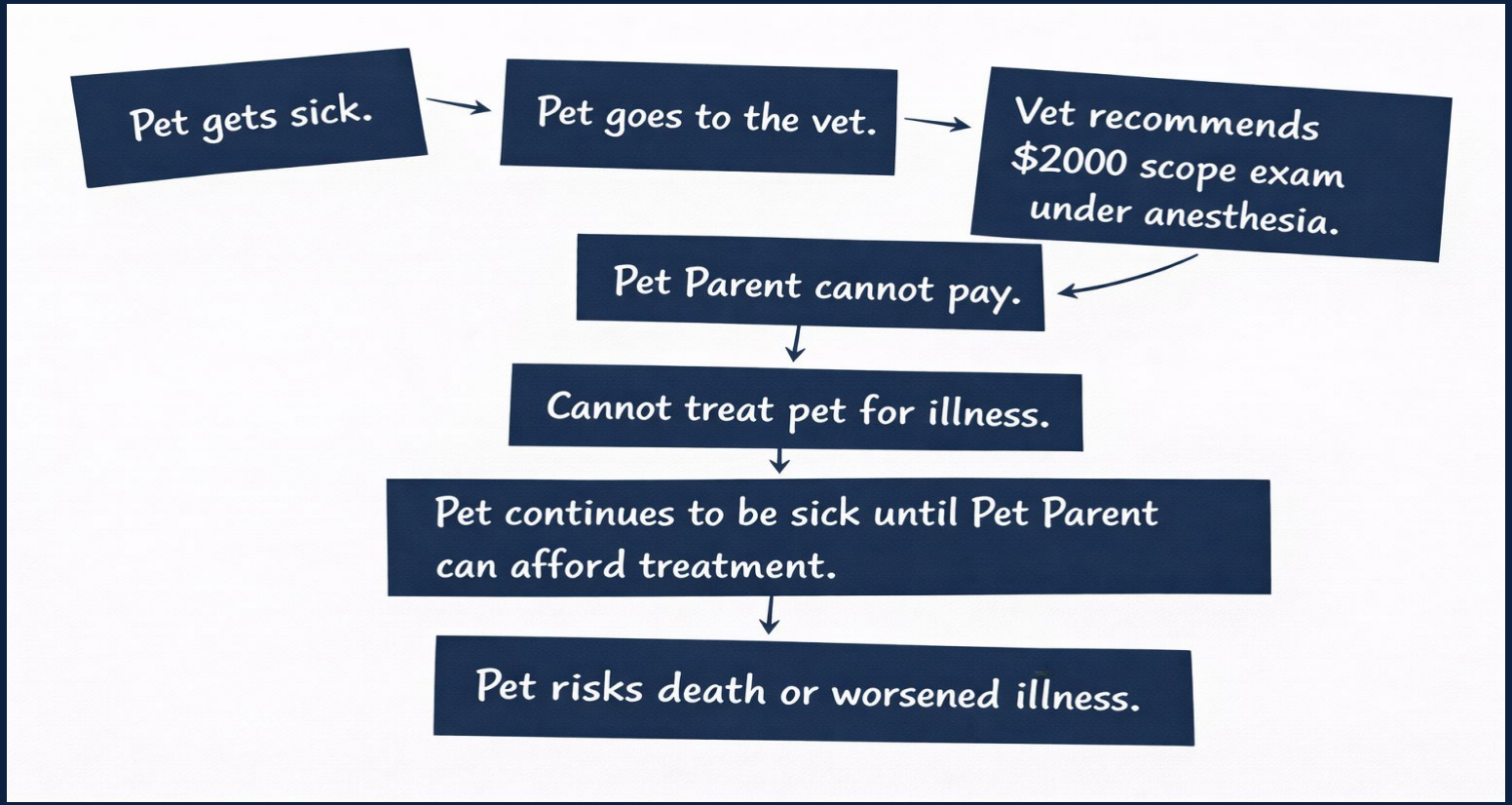
- Visits to the vet are expensive, yet often necessary
  - Means to alleviate veterinary costs such as insurance are rare to have and hard to get proper funding out of

- Options available to circumvent this are not widely available
  - Sources of information can be unreliable or hard to find

Current Process Flow



Current Process Flow Example



OUR SOLUTION

REMOVE THE BARRIER

We pay urgent veterinary bills upfront so pet owners can **act immediately**—without financial delay.

ENABLE COMMUNITY SUPPORT

Members launch a 30-day fundraising campaign with a ***simple, shareable link*** for friends, family, and supporters.

DIRECT & FAIR FUNDING

All donations go directly toward the veterinary bill.
Any extra funds are returned to the ***Member****.*

EMPOWER THROUGH EDUCATION

We provide accessible resources on:

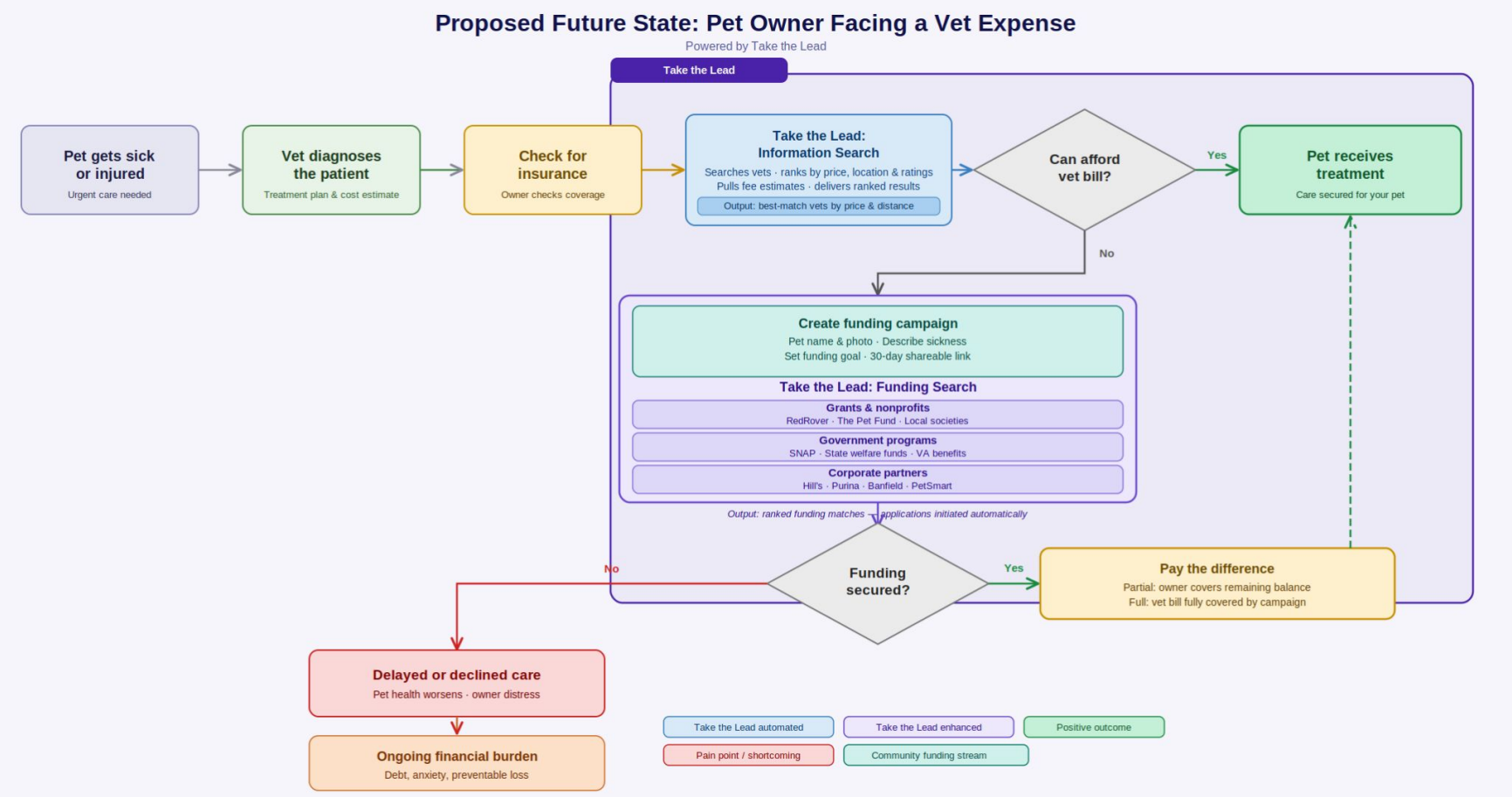
- ***Veterinary costs***
- ***Pet insurance***
- ***Ongoing pet care***

Outcome:

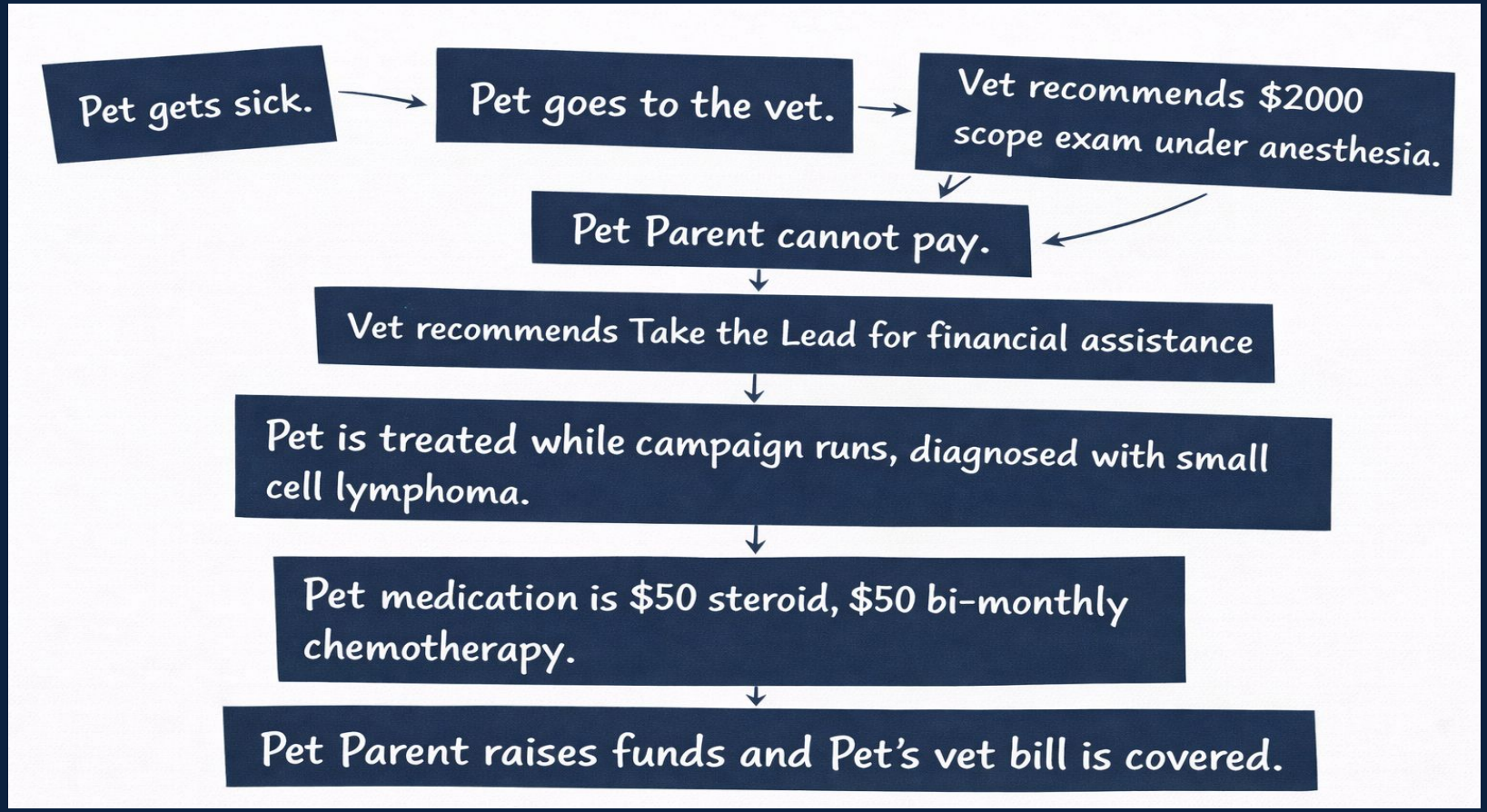
Faster care. Less financial stress. *Better outcomes for pets.*

TAKE THE LEAD

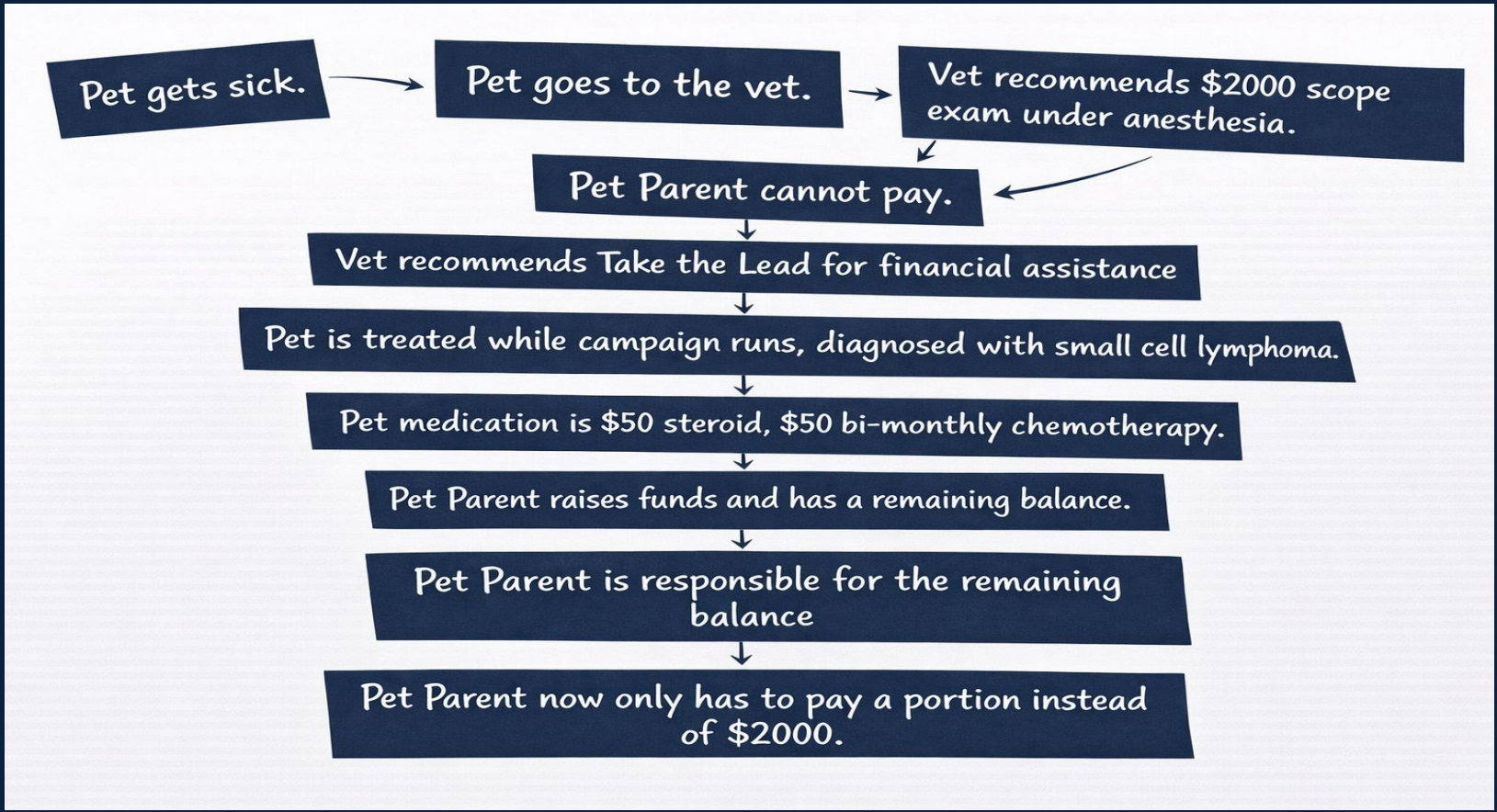
Proposed Solution Flow



Our Solution Example



Our Solution Example Pt 2



Competition Matrix

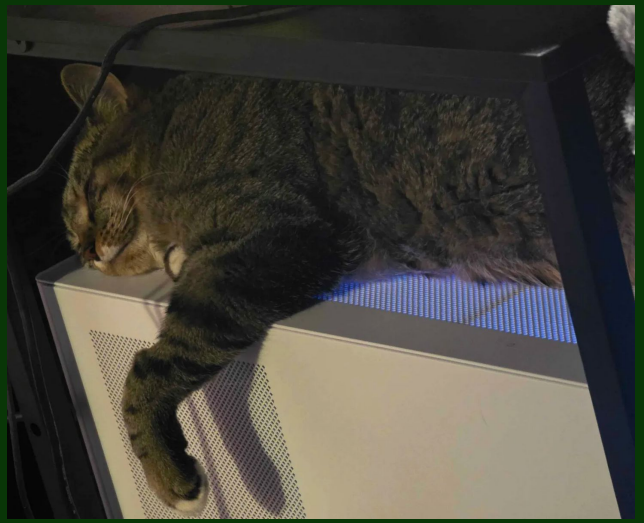
Feature	Take the Lead	Trusted Memories	Frankie's Friends	*The Pet Fund*
Allows users to directly donate to individual pets' medical needs	Yes	Yes		
Provides funding through donation revenue of the service	Initially		Partial (with a $2000 limit)	Yes
Gives easily accessible info and resources for pet owners	Yes			Yes
Gives easily accessible info and resources for veterinarians	Yes			Yes
Uses chat services for users to get more information	Yes			

Why We're Better!

Take The Lead provides many benefits that our competitors do not:

- We ensure 100% of money donated to an individual cause goes to that cause
- There is no hard limit to a Story.
- We do not need to set aside revenue for causes
- The pets directly benefit from the money, we work with professionals to ensure direct contribution

- This includes linked resources from industry professionals and partners to ensure the best quality of information for our customer base
- Interactive chat services will also serve to help users navigate their options



Linked resources will range from Pet-based credit cards, insurance information and other items necessary for a pet's *continued* wellness!

TAKE THE LEAD

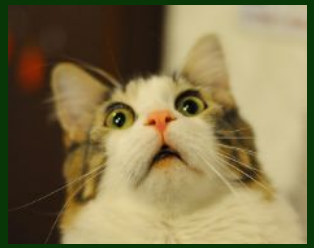
What Our Solution Will Do

- Allow users to facilitate paying for expensive veterinary costs
- A donations platform for friends and family
- Provide extensive educational resources for users
- Provide starting funds for vet treatment if necessary

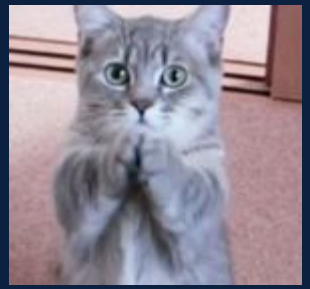
TAKE THE LEAD

What Our Solution Will Not Do

- Direct donations will not incur fees or additional charges
- Story donations will not be used as revenue for our platform
- Does not provide direct physical resources (shelter, food, etc.) to pets
- Will not circumvent the need for veterinary services
- Does not keep excess funds



TAKE THE LEAD



RISKS

Financial Risk

- Campaigns may not raise enough money
- Platform could lose money

Fraud Risk

- Fake campaigns or false vet bills
- Misuse of funds

Legal Risk

- Issues with payments or liability
- Handling sensitive information

Technical Risk

- Website crashes or downtime
- Security vulnerabilities

User Trust Risk

- Users may not trust platform
- Low adoption or donations

External Dependency Risk

- Vet clinics may not cooperate
- Delays in verifying bills

TAKE THE LEAD

MITIGATION

Financial Mitigation

- Set funding limits
- Require partial upfront payment
- Estimate campaign success beforehand

Fraud Mitigation

- Verify vet bills
- User identity checks
- Flag suspicious activity

Legal Mitigation

- Clear terms & agreements
- Follow legal and data regulations

Technical Mitigation

- Reliable servers
- Regular testing & updates
- Strong security practices

User Trust Mitigation

- Transparency
- Reviews & success stories
- Secure payment systems

External Dependency Mitigation

- Partner with clinics
- Standardize verification process
- Incentivize participation

TAKE THE LEAD

Market Size

TAM - Total Addressable Market

90 million US households that own at least one pet.

136 Billion estimated annual pet industry spending

SAM - Serviceable Addressable Market

Between 1 and 3 pet owners face unexpected or emergency vet bills.

$1,000 Average Emergency Vet Cost

SOM - Serviceable Obtainable Market

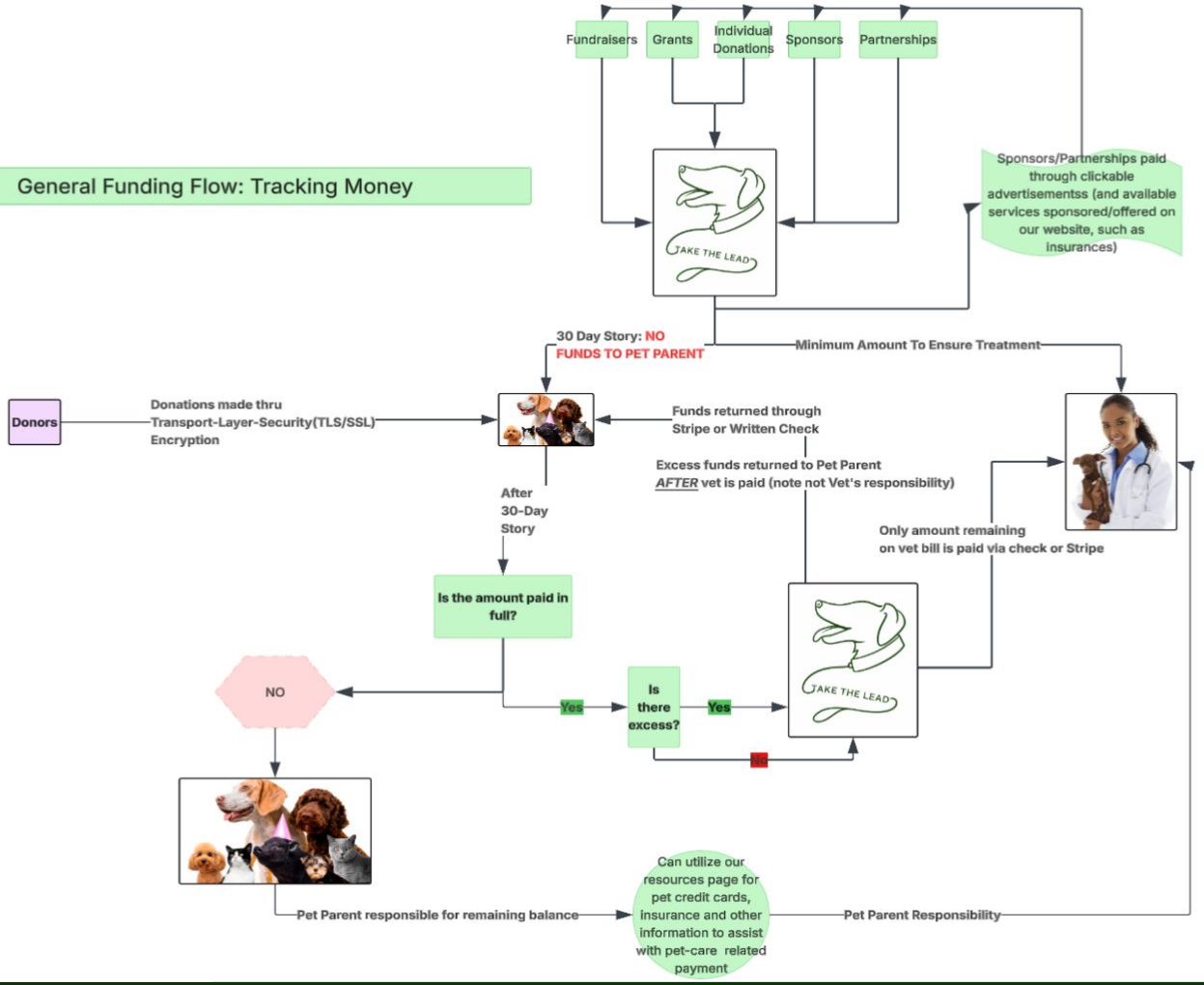
Target Reach - 1 ~ 2% of pet owners

900k - 1.8 Million Users

Monetization - Nonprofit

- Sponsorships/Partnerships
- Charity Fundraisers
- Individual Donations directly to Us
- Merchandise
- Advertisements
- Optional Recursive Funding
- Grants (State and Government) (501c(3) Public)

General Funding Flow: Tracking Money



How We Work: Veterinarian Edition

Verify Pet Parent's vet bill with provided vet information

↓

Confirm payment to vet office for treatment via guaranteed minimum amount (Half of bill below $5000).

↓

Vet office may begin treatment.

↓

Story runs for 30 days (plus grace period, see next slide)

↓

If balance is due,

Funds are distributed via Veterinarian Office's preferred methods (Written check or Bank account via secure payment such as Stripe).

Payments into these accounts may take between 2-5 business days).

Alternative:

If Take the Lead supplied a minimum amount, Take the Lead is paid first and the Veterinarian Office is paid secondary via preferred payment method(s).

TAKE THE LEAD

How Do We Work: Consumer Edition

1. Confirm vet Details
2. Pet is treated immediately
3. Pet parent has 48 hours to Create a story and it runs for 30 days
4. Story is shared online via link to TTL
5. Donors can make secure & anonymous donations (TLS/SSL Encryption)
6. Story Ends after the 30 days

If TTL minimum funding, TTL is paid first

Vet Paid

ONLY if excess: Pet Parents are given remainder (Or Receipt if paid)

Pet Parent to pay reminder

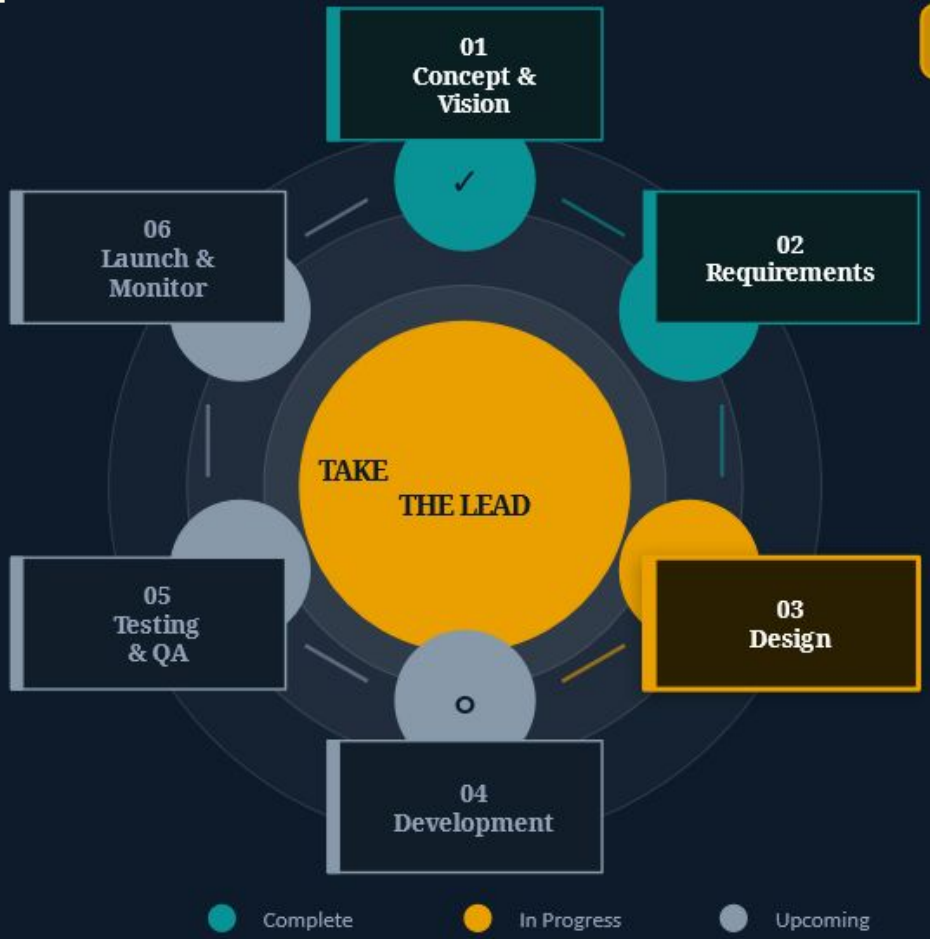
Pet Parent gets receipt of donor names only

TAKE THE LEAD

Development Model -
AGILE ROADMAP

Take The Lead · Web Application

● DESIGN IN PROGRESS



Development Tools

- ## Visual Studio Code

We will mainly use VSCode for programming and updating features and components of the website.

- ## SvelteKit 5+

A framework for development of the consumer-facing website.

- ## PostgreSQL

A query language database for managing user accounts and keeping references to veterinarians.

- ## LLM Assistant

With an LLM that we have access to, we will use it for parsing veterinary data and user data to create our assisted searches.

- ## Git

We will be using Git for version control of the project.

- ## Node.js

Node will be used for programming the website to make it reactive and functional on any platform with a browser.

Section

Major Functional Components

SvelteKit 5

SvelteKit will be used as the main framework for the development of the website. With collaboration, it is easy to control which pages are where and how to handle the flow on the server-side in one repo.

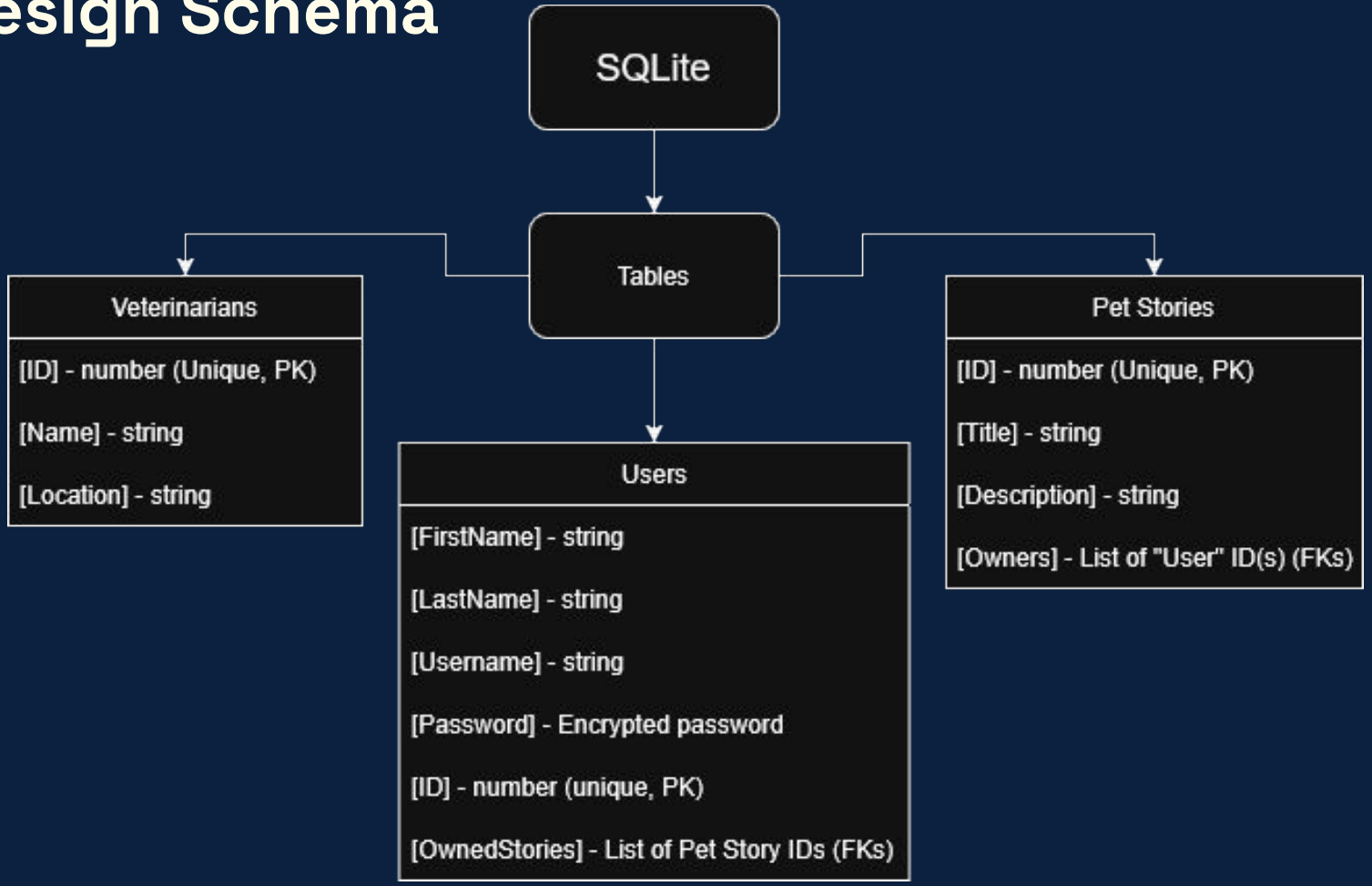
CI Deployment Service

As our project will be Git version controlled, we will need a service such as Render or an on-prem Kubernetes cluster to deploy and maintain the website given updates.

LLM

The LLM is the foundational part here where the LLM is required for processing data and taking user input for the correct output.

Design Schema



Placeholder; Algorithms

Core Platform Features

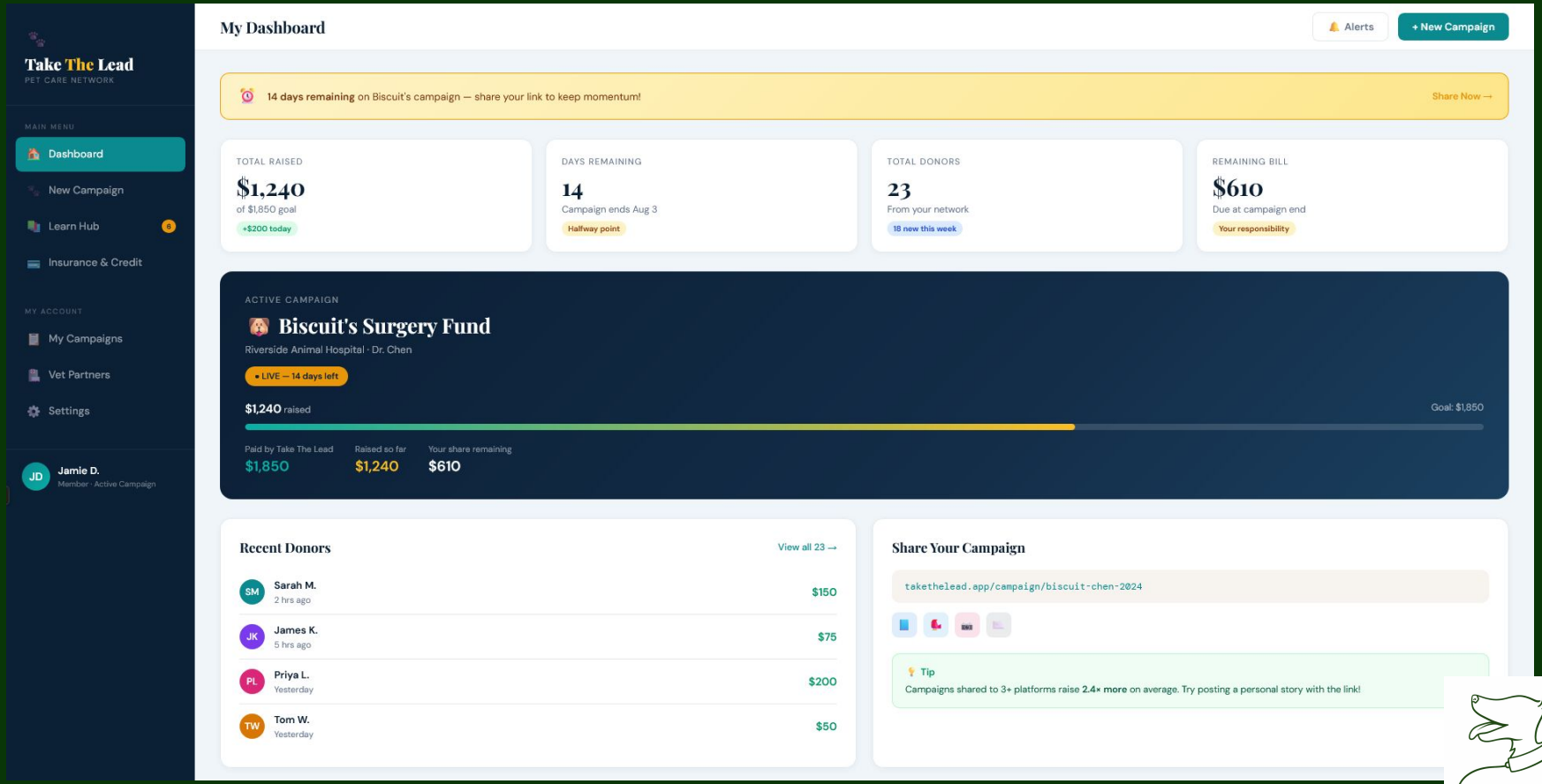
FEATURES	PROTOTYPE	RWP
User Accounts	No MFA Simple login No Verification	OAuth Verification via Email Verification via Phone Detailed Login
Pet Story Creation	Basic Text Limited/No Photos Text Limit	Full Story Builder Interactive Donation Button Video/Photo Uploads Moderation/Tracking
Donation System	Simulated Donations No Secure Payment No Receipts	Protected Payment (Stripe) Receipts Tracking
Fund Distribution	Not Implemented Decoy/Design Function Only	Direct to Take the Lead Direct to Vet Direct to Member
Notification System	None	Email Optional SMS updates

Safety and Compliance

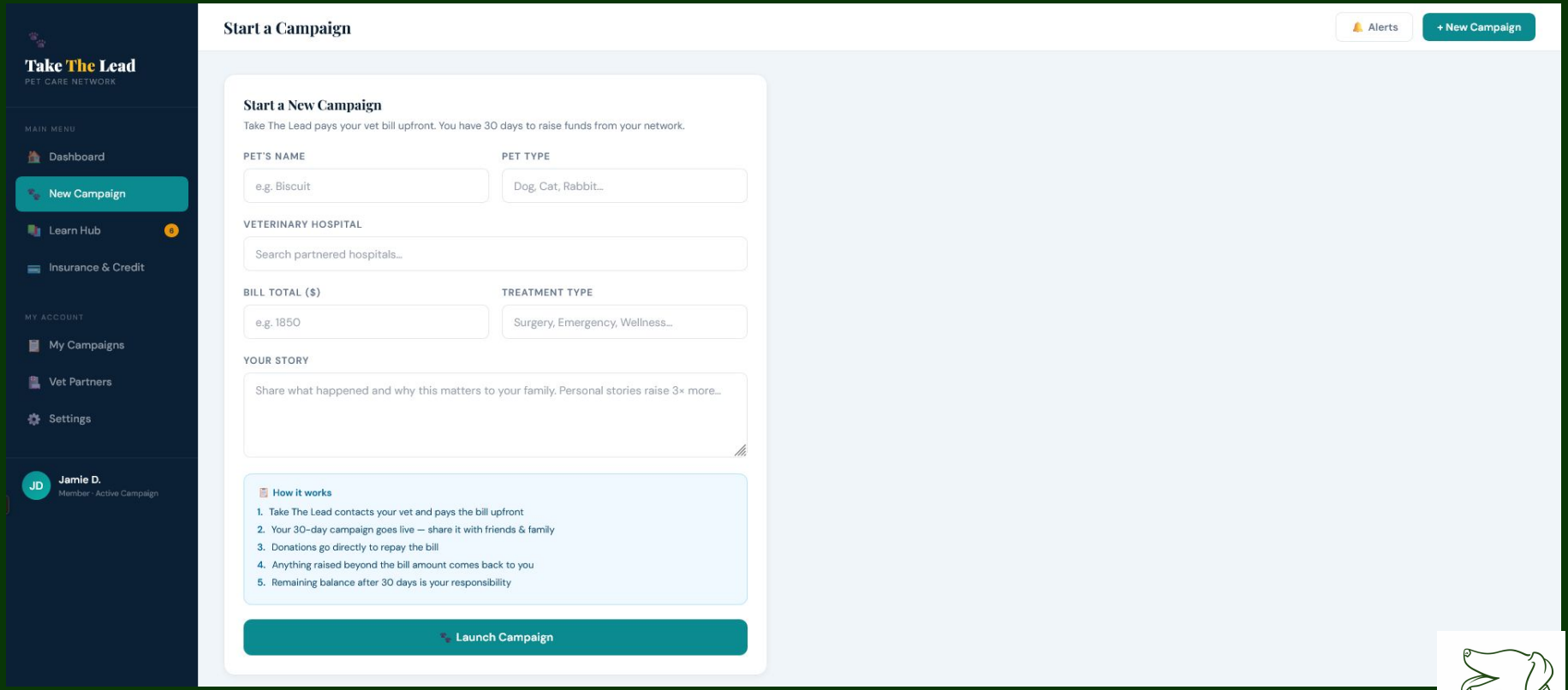
FEATURES	PROTOTYPE	RWP
Payment Security	Not implemented	HTTPS encryption
Verification of User Story	Physically call office to confirm user credentials (address/phone/name)	Verified Pets Connected Pet/Hospitals Fraud Prevention Expedited Approval
Verification of User/Veterinarian	Simulated Donations No Secure Payment No Receipts	Protected Payment (Stripe) Receipts Tracking
Fund Distribution	Not Implemented Decoy/Design Function Only	Direct to Take the Lead Direct to Vet Direct to Member
Notification System	None	Email Optional SMS updates

Placeholder; Deliverables

GUI Mockup


Take The Lead
PET CARE NETWORK
MAIN MENU
Dashboard
New Campaign
Learn Hub
6
Insurance & Credit
MY ACCOUNT
My Campaigns
Vet Partners
Settings
JD
Jamie D.
Member · Active Campaign
My Dashboard
Alerts
+ New Campaign
14 days remaining on Biscuit's campaign — share your link to keep momentum!
Share Now →
TOTAL RAISED
$1,240
of $1,850 goal
+$200 today
DAYS REMAINING
14
Campaign ends Aug 3
Halfway point
TOTAL DONORS
23
From your network
18 new this week
REMAINING BILL
$610
Due at campaign end
Your responsibility
ACTIVE CAMPAIGN
Biscuit's Surgery Fund
Riverside Animal Hospital · Dr. Chen
• LIVE — 14 days left
$1,240 raised
Goal: $1,850
Paid by Take The Lead
$1,850
Raised so far
$1,240
Your share remaining
$610
Recent Donors
View all 23 →
SM
Sarah M.
2 hrs ago
$150
JK
James K.
5 hrs ago
$75
PL
Priya L.
Yesterday
$200
TW
Tom W.
Yesterday
$50
Share Your Campaign
takethelead.app/campaign/biscuit-chen-2024
Tip
Campaigns shared to 3+ platforms raise 2.4× more on average. Try posting a personal story with the link!

GUI Mockup



TAKE THE LEAD

GUI Mockup

Take The Lead
PET CARE NETWORK

MAIN MENU
- Dashboard
- New Campaign
- Learn Hub 6
- Insurance & Credit

MY ACCOUNT
- My Campaigns
- Vet Partners
- Settings

JD Jamie D.
Member · Active Campaign

Learn Hub

Alerts | + New Campaign

All | Insurance | Vet staff | Financing | Prevention

~400%
Veterinarians are up to 4× more likely to die by suicide than the general population*
*AVMA & British Vet Association research

$140K
Average vet school debt — with a median starting salary of $80K. The math is brutal.

60%
Of emergency vet visits could be avoided with annual wellness checkups.

Learning Resources

INSURANCE
Why Pet Insurance Changes Everything
Preventive care plans that reduce emergency costs by up to 60%. Learn what's covered.
Read article →

VET STAFF
The Hidden Crisis Behind Your Vet's Smile
Veterinary professionals face burnout and debt. Understanding their reality builds compassion.
Read article →

PET CREDIT
Pet-Only Cards: Better Rates, Less Stress
Specialized credit options with lower APR and deferred interest for qualified medical care.
Read article →

PREVENTION
The True Cost of Skipping Checkups
Annual wellness visits catch problems early — and cost 80% less than emergency treatment.
Read article →

FINANCE
Vet Debt: A $140K Reality
Average vet school debt vs. median salary. Why pricing isn't personal — it's survival.
Read article →

PARTNERSHIPS
Hospital Agreements Explained
How Take The Lead works with your vet to guarantee care before funding is complete.
Read article →

TAKE THE LEAD

References

Treasured Memories

"Treasured Memories Community Funding." *Treasured Memories Community Funding*, 2019, www.tmcfunding.com. Accessed 12 Mar. 2026.

Frankie's Friends

"Welcome." *Frankie's Friends*, 2026, www.frankiesfriends.org. Accessed 12 Mar. 2026.

The Pet Fund

"Home." *The Pet Fund*, 2003-2026, www.thepetfund.com. Accessed 12 Mar. 2026.

Facts + Statistics

"Facts + Statistics: Pet Ownership and Insurance." *Insurance Information Institute*, 2026, www.iii.org/fact-statistic/facts-statistics-pet-ownership-and-insurance. Accessed 12 Mar. 2026.

CDC

Nett, Randall J., et al. "Suicide Risk for Veterinarians and Veterinary Technicians." *NIOSH Science Bulletin*, Centers for Disease Control and Prevention, 4 Sept. 2019, www.cdc.gov/niosh/bulletin/2019/suicide-veterinarians.html. Accessed 12 Mar. 2026.

AMVA

"Home." *American Veterinary Medical Association*, www.avma.org/. Accessed 12 Mar. 2026.

References (Cont.)

- **Not One More Vet**

  "Veterinary Profession Mental Wellness." *Not One More Vet*, 2025, www.nomv.org/. Accessed 12 Mar. 2026.

- **X**

  x

- **X**

  x

- **X**

  x

- **X**

  x

- **X**

  x

Placeholder; Appendix/ User Stories